
Deep-learning pansharpening

Launcher: `Toolbox/pansharpening_cnn.bat`

This optional tool requires `env_pnn` with compatible PyTorch setup. See main README.

Purpose

The tool aligns matching multispectral and panchromatic rasters, runs one of six Z-PNN-family networks, and exports both the native MAT result and a georeferenced GeoTIFF.

Inputs and parameters

Option	Meaning and behaviour
Multispectral (MS/ML) raster	WV3/WV2 input must have 8 bands; GE1 must have 4. It must overlap the PAN raster, use the same CRS, and be north-up.
Panchromatic (PAN) raster	Matching single-band raster. The common intersection is snapped inward to complete MS pixels, and aligned PAN dimensions are forced to four times the MS width and height.
Pansharpening algorithm	Z-PNN, A-PNN-TA-FR, Z-PanNet, PanNet-TA-FR, Z-DRPNN, or DRPNN-TA-FR. Default: Z-PNN. Z-prefixed methods use zero-shot adaptation; TA-FR variants use target-adaptive full-resolution training according to the bundled implementation.
Sensor	WV3, WV2, or GE1. Default: WV3. It selects expected band count and pretrained weights; it does not inspect sensor metadata automatically.
Fine-tuning epochs	Non-negative integer. Default: 1. 0 runs the loaded pretrained model without fine-tuning. In Z-PNN, 1 is a special shortcut interpreted as 100 epochs for a Z- method and 2,000 for a TA-FR method.

Option	Meaning and behaviour
Use CPU	Off by default. Enable when CUDA is unavailable. Processing will be considerably slower.
Enable co-registration loss	On by default. Adds the bundled model's co-registration term during adaptation. Disable only for an intentional comparison or when troubleshooting that term.
Output folder	Blank creates pansharp_d1 beside the MS raster. A manually selected folder must already exist.

Outputs

- `<MS-name>_<algorithm>.mat` — native Z-PNN output using hyphens in the algorithm name.
- `<MS-name>_<algorithm_with_underscores>.tif` — georeferenced, tiled, DEFLATE-compressed UInt16 GeoTIFF on the aligned PAN grid.

GeoTIFF values are clipped to the UInt16 range 0–65535. Temporary aligned rasters and input MAT data are removed. Source rasters are never modified. Existing outputs may be replaced because this GUI has no overwrite confirmation.

Quality checks

Check alignment on sharp roads and field edges, look for halos or duplicated boundaries, compare each band with the original MS data, and examine output histograms for clipping at 0 or 65535. Network detail is inferred, not directly measured, so do not treat small sharpened forms as independent archaeological evidence.